

Intercity Delivery System Design

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Omitted.

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1. Model Specification

Table 1: Notation List

Notation	Explanation
Parameters	
$\mathcal{T}$	discretized time periods
$\mathcal{D}^+, \mathcal{D}^-$	order categorization set
d_l	number of type-l orders
s_l	earliest start time of type-l orders
e_l	latest completion time of type-l orders
N^i	number of available vehicles for city $i \in \{1, 2\}$
$\hat{N}^i$	number of available automated vehicles
M	capacity of manually driven vehicles
$\hat{M}$	capacity of automated vehicles
c	unit driving cost for manually driven vehicles
$\hat{c}$	unit driving cost for automated vehicles
δ_l	unit penalty cost for lost demand
τ	driving time between cities 1 and 2
$f^i(\cdot)$	service time function for city $i \in \{1, 2\}$
t_0	duration of a single period
$x_{ij}^{k\uparrow} \in \mathbb{Z}^+, \forall i, j \in \mathcal{T}, k \in \{1, 2\}, \uparrow \in \{+, -\}$	number of manually driven vehicles performing the corresponding delivery task during time interval $[i, j]$
$y_{ij}^{\uparrow} \in \mathbb{Z}^+, \forall i, j \in \mathcal{T}, \uparrow \in \{+, -\}$	number of automated vehicles traveling between the two cities from periods i to j
$g_{ij}^l \in \mathbb{R}^+, \forall i, j \in \mathcal{T}, l \in \mathcal{D}$	intercity freight volume carried by manually driven vehicles
$\hat{g}_{ij}^l \in \mathbb{R}^+$	intercity freight volume carried by automated vehicles
z_l	unserved freight volume

We label the two cities as 1 and 2, and use the superscripts + and - to denote flows from 1 to 2 and from 2 to 1, respectively. Define $\hat{\mathcal{A}} := \{(i, j) : i, j \in \mathcal{T}, i + \tau = j\}$ as the set of all possible period pairs

for $y_{ij}^\dagger$. Similarly, $\mathcal{A}^k := \{(i, j) : j \leq i + f^k(M)\}, \forall k \in \{1, 2\}$ is defined as the set of all possible period pairs for $x_{ij}^{k\dagger}$. We consider that $f^i(\cdot), \forall i \in \{1, 2\}$, is concave, e.g., $f^i(\lambda) = a^i\lambda + b^i\sqrt{\lambda}$ (BHH approximation). Define $\mathcal{D} = \mathcal{D}^+ \cup \mathcal{D}^-$. In addition, we define $\mathcal{S}^k(t) := \{(i, j) : (i, j) \in \mathcal{A}^k, i \leq t < j\}, \forall t \in \mathcal{T}, k \in \{1, 2\}$. Similarly, $\hat{\mathcal{S}}(t) := \{(i, j) : (i, j) \in \hat{\mathcal{A}}, i \leq t < j\}$.

$$\min_{\mathbf{x}, \mathbf{y}, \mathbf{g}, \hat{\mathbf{g}}, \mathbf{z}} \sum_{l \in \mathcal{D}} \delta_l z_l + \sum_{k \in \{1, 2\}} \sum_{\dagger \in \{+, -\}} \sum_{(i, j) \in \mathcal{A}^\dagger} x_{ij}^{k\dagger} (j - i) c t_0 + \sum_{(i, j) \in \hat{\mathcal{A}}} \sum_{\dagger \in \{+, -\}} y_{ij}^\dagger \hat{c} \tau t_0 \quad (1)$$

$$\text{s.t.} \quad \sum_{\dagger \in \{+, -\}} \sum_{(i, j) \in \mathcal{S}^k(t)} x_{ij}^{k\dagger} \leq N^k, \quad \forall k \in \{1, 2\}, t \in \mathcal{T}, \quad (2)$$

$$\sum_{\dagger \in \{+, -\}} \sum_{(i, j) \in \hat{\mathcal{S}}(t)} y_{ij}^\dagger \leq \hat{N}^1 + \hat{N}^2, \quad \forall t \in \mathcal{T}, \quad (3)$$

$$\sum_{(i, j) : (i, j) \in \hat{\mathcal{A}}, i \leq t} y_{ij}^+ - \sum_{(i, j) : (i, j) \in \hat{\mathcal{A}}, j \leq t} y_{ij}^- + \hat{N}^1 \geq 0, \quad \forall t \in \mathcal{T}, \quad (4)$$

$$\sum_{(i, j) : (i, j) \in \hat{\mathcal{A}}, i \leq t} y_{ij}^- - \sum_{(i, j) : (i, j) \in \hat{\mathcal{A}}, j \leq t} y_{ij}^+ + \hat{N}^2 \geq 0, \quad \forall t \in \mathcal{T}, \quad (5)$$

$$\sum_{l \in \mathcal{D}^\dagger} g_{ij}^l \leq x_{ij}^{k\dagger} \cdot (f^k)^{-1}[(j - i)t_0], \quad \forall (i, j) \in \mathcal{A}^k, \dagger \in \{+, -\}, k \in \{1, 2\}, \quad (6)$$

$$\hat{g}_{ij}^l \leq y_{ij}^\dagger \hat{M}, \quad \forall \dagger \in \{+, -\}, l \in \mathcal{D}^\dagger, (i, j) \in \hat{\mathcal{A}}, \quad (7)$$

$$g_{ij}^{l\dagger} = 0, \quad \forall (i, j, l, \dagger) \in \mathcal{E}, \quad (8)$$

$$\sum_{(i', j') : (i', j') \in \hat{\mathcal{A}}, i' \leq i} \hat{g}_{i'j'}^l \leq \sum_{(i', j') : (i', j') \in \mathcal{A}^{\eta\dagger}, j' \leq i} g_{i'j'}^l, \quad \forall i \in \mathcal{T}, \dagger \in \{+, -\}, l \in \mathcal{D}^\dagger, \quad (9)$$

$$\sum_{(i', j') : (i', j') \in \hat{\mathcal{A}}, j' \leq i} \hat{g}_{i'j'}^l \geq \sum_{(i', j') : (i', j') \in \mathcal{A}^{\bar{\eta}\dagger}, i' \leq i} g_{i'j'}^l, \quad \forall i \in \mathcal{T}, \dagger \in \{+, -\}, l \in \mathcal{D}^\dagger, \quad (10)$$

$$d_l = z_l + \sum_{(i, j) \in \mathcal{A}^k} g_{ij}^l, \quad \forall k \in \{1, 2\}, l \in \mathcal{D}, \quad (11)$$

where $\mathcal{E} := \{(i, j, l, +) : (i, j) \in \mathcal{A}^1, s_l > i \text{ or } (i, j) \in \mathcal{A}^2, e_l < j\} \cup \{(i, j, l, -) : (i, j) \in \mathcal{A}^2, s_l > i \text{ or } (i, j) \in \mathcal{A}^1, e_l < j\}$ denotes the set of time pairs that violate the time-window constraints, $\eta_\dagger = \begin{cases} 1, & \text{if } \dagger = + \\ 2, & \text{otherwise} \end{cases}$, and $\bar{\eta}_\dagger = \begin{cases} 2, & \text{if } \dagger = + \\ 1, & \text{otherwise} \end{cases}$. Constraints (2)–(5) restrict the fleet size. Constraints (6)–(7) ensure that the load assigned to each vehicle type does not exceed its capacity. Constraint (8) imposes service time-window constraints on passengers. Constraints (9)–(10) ensure flow conservation of passenger demand at transfer nodes. Finally, Constraint (11) captures the unserved passenger volume.